Senior C++ *Blockchain* Engineer — *Web3* Fintech & Payments

Location: Remote / Poland (International Team)
Employment Type: Full-time / Part-time / Contract
Reports To: Head of *Blockchain* Engineering / CTO
Level: Senior Engineer (Hands-on Technical Role)

About the Company

Plaid is a financial technology company that provides infrastructure connecting financial applications with banks and financial institutions. Its technology supports secure financial data access, account verification, identity verification, and payment services.

Separately, we are planning to establish a new software development company in Poland, focusing on fintech, payment infrastructure, and *blockchain*-based financial applications.

Our planned products cover real estate marketplaces, gaming, sports betting, and *Web3* social finance platforms.

We aim to combine traditional financial technologies with *blockchain* infrastructure to support secure digital payments, *wallet* integrations, transaction processing, settlement systems, and decentralized financial services.

We are building an international engineering team and looking for experienced engineers who can contribute to our infrastructure and architecture from an early stage.

About the Role

We are looking for an experienced **Senior C++ *Blockchain* Engineer** to design, develop, and maintain high-performance *blockchain* infrastructure and distributed systems.

This is a hands-on engineering position focused on **modern C++, *blockchain* protocols, networking, transaction processing, and systems-level performance optimization**.

You will work closely with *blockchain* protocol engineers, backend developers, infrastructure engineers, and security specialists to develop reliable components supporting *blockchain* networks and financial applications.

Your responsibilities will include developing *blockchain* node components, implementing transaction-processing services, optimizing network communication, and improving the performance and reliability of distributed *blockchain* infrastructure.

You may also contribute to *blockchain* clients, consensus-related components, *crypto*graphic integrations, and cross-chain infrastructure, depending on project requirements.

We are looking for someone with s*tron*g C++ engineering fundamentals who understands distributed systems and has practical experience working with *blockchain* protocols or *cryptocurrency* infrastructure.

This is not a purely DevOps, frontend, or *Solidity smart contract* development position. The primary focus is building and maintaining performant *blockchain* software using C++.

What You'll Do

C++ Systems Development

- Design, develop, and maintain high-performance services and *blockchain* infrastructure using modern C++.
- Write efficient, secure, and maintainable C++17/20 code.
- Develop concurrent and multithreaded components for transaction processing, networking, and data synchronization.
- Optimize memory allocation, CPU utilization, disk I/O, and network performance.
- Build reusable libraries and modular software components.
- Implement asynchronous programming, thread-safe data structures, and efficient task-processing systems.
- Investigate race conditions, deadlocks, memory corruption, and performance bottlenecks.
- Apply appropriate memory management techniques, including RAII, smart pointers, and move semantics.
- Improve software stability through profiling, benchmarking, testing, and debugging.
- Participate in architecture discussions and contribute to critical technical decisions.

Blockchain Protocol & Node Engineering

- Develop and maintain *blockchain* node software and related infrastructure components.
- Work with *blockchain* transaction structures, blocks, state management, and validation mechanisms.
- Implement or integrate transaction validation, block processing, and state synchronization.
- Develop components supporting *blockchain* networking and peer-to-peer communication.
- Improve node performance, synchronization speed, and operational reliability.

- Investigate *blockchain* reorganizations, transaction propagation delays, and synchronization failures.
- Work with *blockchain* data structures such as Merkle trees, transaction pools, and authenticated state representations.
- Collaborate with protocol engineers on *blockchain* client upgrades and network compatibility.
- Contribute to consensus-related functionality where required and appropriate to the selected *blockchain* architecture.
- Support integration with *Bitcoin*, *Ethereum*-compatible, or other *blockchain* networks based on product requirements.

Distributed Systems & Networking

- Design and build fault-tolerant distributed services.
- Develop high-performance peer-to-peer networking components.
- Work with TCP/IP, asynchronous I/O, RPC, and messaging protocols.
- Implement reliable communication between *blockchain* nodes and application services.
- Design systems that handle network partitions, retries, message duplication, and temporary failures.
- Optimize service latency, throughput, and resource utilization.
- Develop internal APIs and interfaces supporting *blockchain* infrastructure.
- Collaborate with backend engineers on distributed transaction-processing workflows.
- Investigate performance and consistency issues across distributed components.

Transaction Processing & Financial Infrastructure

- Build infrastructure supporting digital asset transaction processing and *blockchain*-connected financial applications.
- Develop services for transaction creation, submission, verification, and confirmation tracking.
- Integrate *blockchain* services with backend payment and settlement systems.
- Support accurate transaction state management and reconciliation between on-chain and off-chain systems.
- Implement reliable handling for failed transactions, duplicate requests, delayed confirmations, and recovery scenarios.
- Develop monitoring and processing components for digital asset transfers, deposits, and withdrawals.
- Collaborate with security engineers on secure transaction authorization and signing integrations.
- Optimize transaction-processing performance while preserving correctness and reliability.
- Support infrastructure used by *wallet* services, payment platforms, and decentralized applications.

*Crypto*graphy & *Blockchain* Security

- Integrate established *crypto*graphic libraries and primitives into *blockchain* infrastructure.
- Work with digital signatures, hashing algorithms, Merkle proofs, and transaction verification.
- Understand *blockchain* key management and secure signing requirements.
- Implement secure handling of sensitive data and *crypto*graphic materials.
- Review C++ code for memory safety, input validation, integer overflow, and concurrency vulnerabilities.
- Collaborate with security engineers to evaluate threats to *blockchain* infrastructure.
- Follow established *crypto*graphic standards and avoid unnecessary custom *crypto*graphic implementations.
- Participate in security reviews, testing, and remediation of identified vulnerabilities.

Blockchain Data & Infrastructure Integration

- Develop services that process *blockchain* blocks, transactions, and contract events.
- Build components supporting *blockchain* in*dex*ing and historical data synchronization.
- Integrate *blockchain* infrastructure with PostgreSQL, RocksDB, LevelDB, or similar storage systems.
- Optimize storage access, serialization, caching, and data retrieval.
- Develop internal APIs for *blockchain* data and transaction information.
- Handle historical backfills and recovery from incomplete synchronization.
- Work with backend teams to expose reliable *blockchain* data to financial applications.
- Support infrastructure monitoring and observability.

Testing, Performance & Reliability

- Develop unit, integration, and end-to-end tests for C++ components.
- Use Google Test, Catch2, or comparable testing frameworks.
- Build performance benchmarks for critical system components.
- Use sanitizers, profiling tools, and debuggers to identify reliability issues.
- Work with AddressSanitizer, ThreadSanitizer, Valgrind, GDB, or equivalent tooling.
- Test behavior under network instability, node failures, unexpected inputs, and heavy transaction loads.
- Contribute to continuous integration and automated testing pipelines.
- Review pull requests and maintain code quality standards.
- Investigate production incidents and implement reliable long-term fixes.
- Document architectural decisions, performance improvements, and important system behavior.

Infrastructure & Collaboration

- Develop and maintain C++ build systems using CMake or similar tooling.

- Work with Git, Docker, Linux, and CI/CD pipelines.
- Collaborate with DevOps engineers on deployments and operational requirements.
- Support integration testing across *blockchain* nodes, backend services, and financial applications.
- Participate in technical planning and architecture reviews.
- Mentor less experienced engineers through code reviews and technical discussions.
- Work with globally distributed engineering teams.

Required Skills & Experience

- 5+ years of professional software engineering experience.
- 3+ years of substantial professional C++ development experience.
- S*trong* proficiency in **modern C++ (C++17/20)**.
- Practical experience with *blockchain* protocols, *cryptocurrency* infrastructure, or distributed *ledger* systems.
- S*trong* understanding of distributed systems, networking, and concurrent programming.
- Experience developing multithreaded, asynchronous, or performance-sensitive applications.
- Knowledge of memory management, smart pointers, RAII, and C++ concurrency primitives.
- Experience with Linux-based development and debugging.
- Familiarity with *blockchain* fundamentals, including transactions, blocks, consensus, finality, and chain reorganizations.
- Understanding of peer-to-peer networking and *blockchain* node communication.
- Experience working with RPC interfaces, networking libraries, or distributed backend services.
- Familiarity with *crypto*graphic signatures, hashing, and *blockchain* data structures.
- Experience with CMake and C++ build tooling.
- Knowledge of Git, automated testing, and code review practices.
- Experience using debugging and profiling tools to investigate system issues.
- S*trong* understanding of software architecture and performance optimization.
- Ability to independently design, develop, test, and maintain complex systems components.
- S*trong* problem-solving skills and experience working on production software.
- Professional English communication skills.

Nice to Have (Differentiators)

- Experience contributing to *Bitcoin* Core, *Bitcoin*-derived clients, *Ethereum* clients, or other open-source *blockchain* implementations.
- Familiarity with *Bitcoin* transaction validation, UTXO models, mempool behavior, and networking.
- Knowledge of *Ethereum* execution clients, *EVM* internals, or Layer 2 infrastructure.

- Experience working with *blockchain* consensus and validator software.
- Familiarity with libp2p or other peer-to-peer networking libraries.
- Experience with Boost.Asio, gRPC, Protobuf, or similar networking and serialization technologies.
- Experience developing low-latency financial or transaction-processing infrastructure.
- Knowledge of RocksDB, LevelDB, or high-performance embedded storage engines.
- Familiarity with *Rust* or Go for *blockchain* infrastructure development.
- Experience with *crypto*graphic libraries such as OpenSSL, libsodium, or secp256k1.
- Knowledge of advanced concurrency techniques, lock-free structures, or efficient memory allocators.
- Experience with cross-chain bridges, interoperability infrastructure, or *blockchain* in*dex*ing.
- Familiarity with *blockchain wallet*s, custody infrastructure, HSMs, or secure key management.
- Knowledge of zero-knowledge proof verification or other *crypto*graphic infrastructure.
- Experience working with Docker, Kubernetes, and cloud infrastructure.
- Understanding of secure software development for financial systems.
- Open-source contributions to C++, *blockchain*, *crypto*graphy, or distributed systems projects.
- Experience in fintech, digital asset exchanges, *cryptocurrency* payments, or decentralized finance.

Senior-Level Expectations

As a Senior C++ *Blockchain* Engineer, you will be expected to:

- Independently design and implement complex C++ components.
- Take ownership of important systems from technical design through production maintenance.
- Make sound architecture decisions involving performance, concurrency, security, and reliability.
- Understand how *blockchain* protocols interact with backend and financial infrastructure.
- Identify performance bottlenecks and resolve difficult systems-level problems.
- Design software that remains reliable under unexpected failures and heavy workloads.
- Write well-tested, maintainable, and secure code.
- Contribute to code reviews and engineering standards.
- Collaborate with protocol, backend, security, and DevOps engineers.
- Mentor engineers and share practical systems programming knowledge.
- Investigate complex production issues and implement long-term solutions.
- Remain actively involved in hands-on engineering.

This is a senior individual contributor position. Formal people management is not required, but technical ownership and engineering collaboration are important.

Example Problems You'll Work On

1. High-Performance Transaction Processing

Develop a C++ service that receives, validates, and processes *blockchain* transaction data.

Optimize processing latency and throughput while ensuring correct handling of duplicate transactions, network failures, and unexpected inputs.

2. *Blockchain* Node Integration

Build components that communicate with *blockchain* nodes through RPC or peer-to-peer interfaces.

Implement efficient synchronization, status monitoring, and recovery from connection failures.

3. Transaction Confirmation Tracking

Develop infrastructure that monitors *blockchain* transactions from submission through confirmation.

Handle changing confirmation states, chain reorganizations, and delayed or failed transactions.

4. Distributed Financial Services

Build C++ services supporting secure communication between *blockchain* infrastructure and financial backend applications.

Design reliable processing and recovery mechanisms for transaction-related workflows.

5. *Blockchain* Data Processing

Develop a high-performance pipeline that processes blocks and transaction data.

Optimize serialization, database access, caching, and synchronization for efficient data retrieval.

6. Peer-to-Peer Networking

Implement and improve networking components used by *blockchain* nodes or related distributed services.

Address message propagation, connection management, concurrency, and network reliability.

7. *Crypto*graphic Transaction Verification

Integrate established *crypto*graphic libraries to verify *blockchain* signatures and transaction proofs.

Ensure correct input handling, efficient verification, and secure implementation.

8. Performance & Reliability Optimization

Investigate a resource-intensive *blockchain* component using profilers, benchmarks, and debugging tools.

Identify bottlenecks, improve performance, and add automated tests to prevent regressions.

Why This Role Matters

As we establish our planned software development operations in Poland, reliable and high-performance *blockchain* infrastructure will be an important foundation for our fintech and *Web3* applications.

Our products will require secure transaction processing, efficient *blockchain* network communication, reliable data synchronization, and dependable integration with financial backend systems.

The Senior C++ *Blockchain* Engineer will help develop the systems-level components supporting these capabilities.

This role offers meaningful ownership of *blockchain* infrastructure, distributed systems, and performance-critical software.

You will collaborate with experienced engineers, contribute directly to important technical architecture, and help establish s*tron*g engineering practices as our international team grows.

We are looking for an engineer who combines **advanced C++ development skills, practical *blockchain* knowledge, distributed systems expertise, and s*tron*g experience building secure, reliable, high-performance software**.